

Solution Requirement

Date	30 September 2026
Team ID	SWTID-2026-8550
Project Title	AI BlogNest API

Solution Requirement

Functional Requirements

FR-01 User registration and login.
FR-02 JWT authentication.
FR-03 RBAC for Admin, Editor, Author and Reader.
FR-04 Blog CRUD operations.
FR-05 Categories and tags.
FR-06 Comment creation and moderation.
FR-07 Media metadata/path handling.
FR-08 Search capabilities.
FR-09 AI blog generation.
FR-10 AI summarization.
FR-11 Input validation and sanitization.
FR-12 Centralized errors, rate limiting and structured logging.

Non-Functional Requirements

Type	Requirement
Security	JWT, RBAC, bcrypt.js, validation, sanitization.
Performance	Efficient database/service execution.
Scalability	Layered modular architecture.
Reliability	Schema validation and controlled access.
Maintainability	MVC and modular services.
Portability	Windows, macOS and Linux.
Usability	Postman/ThunderClient testing.

Integrity	Mongoose schema validation.
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Hardware Requirements

Intel Core i5 8th Gen+ / AMD Ryzen 5 equivalent.

8 GB RAM minimum; 16 GB recommended.

1 GB available storage.

Software Requirements

Windows 10/11, macOS or Linux.

Node.js v16+.

npm v8+.

Express.js.

MongoDB.

Postman.

Visual Studio Code.